DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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INFORMATION ASSURANCE AND SECURITY
(24CS3105)

ALM – PROTOTYPE REVIEW

submitted by

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Abstract

This project presents a practical study and implementation of fundamental Information Assurance and Security (IAS) concepts through various cryptographic algorithms. The prototype includes Caesar Cipher, Playfair Cipher, Hill Cipher, Vigenère Cipher, Simplified Data Encryption Standard (SDES), Advanced Encryption Standard (AES), Stream Cipher (RC4), Diffie–Hellman Key Exchange, and RSA. These algorithms demonstrate different approaches to protecting information during storage and transmission.

The implementations are developed using Python and follow a client-server communication model, where messages are encrypted before transmission and decrypted at the receiving end. Classical cryptographic algorithms are implemented to understand the basic principles of substitution, key-based encryption, and mathematical transformations. Modern cryptographic techniques such as AES and RC4 demonstrate symmetric encryption, while RSA demonstrates asymmetric encryption using public and private keys. Diffie–Hellman demonstrates the secure establishment of a shared secret key between communicating parties. PyCryptodome is used for the algorithms supported by the library.

The prototype provides hands-on understanding of how cryptographic techniques contribute to confidentiality, integrity, authentication, and secure communication, which are essential objectives of Information Assurance and Security. Through the implementation and testing of these algorithms, the project demonstrates their basic working principles and their application in securing communication between networked systems.

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Chapter 1

Cryptographic Algorithms

1.1 Caesar Cipher

1.1.1 Algorithm Definition

The Caesar Cipher is a classical substitution cipher in which each letter of the plaintext is shifted by a fixed number of positions in the alphabet. The same shift key is used for encryption and decryption.

1.1.2 Mathematical Notation

Let P represent the plaintext character, C represent the ciphertext character, and K represent the shift key.

The encryption operation is:

$$C = (P + K) \bmod 26 \tag{1.1}$$

The decryption operation is:

$$P = (C - K) \bmod 26 \tag{1.2}$$

1.1.3 Screenshot

Figure 1.1: Caesar Cipher Client-Server Communication

1.2 Playfair Cipher

1.2.1 Algorithm Definition

The Playfair Cipher is a classical digraph substitution cipher that encrypts two plaintext characters at a time. A 5×5 matrix is generated using a keyword, and character pairs are encrypted according to their positions in the matrix.

1.2.2 Mathematical Notation

For two characters in the same row, each character is replaced by the character immediately to its right, with wrapping at the end of the row.

For two characters in the same column, each character is replaced by the character immediately below it, with wrapping at the bottom of the column.

For characters forming a rectangle, each character is replaced by the character in the same row and the column occupied by the other character.

1.2.3 Screenshot

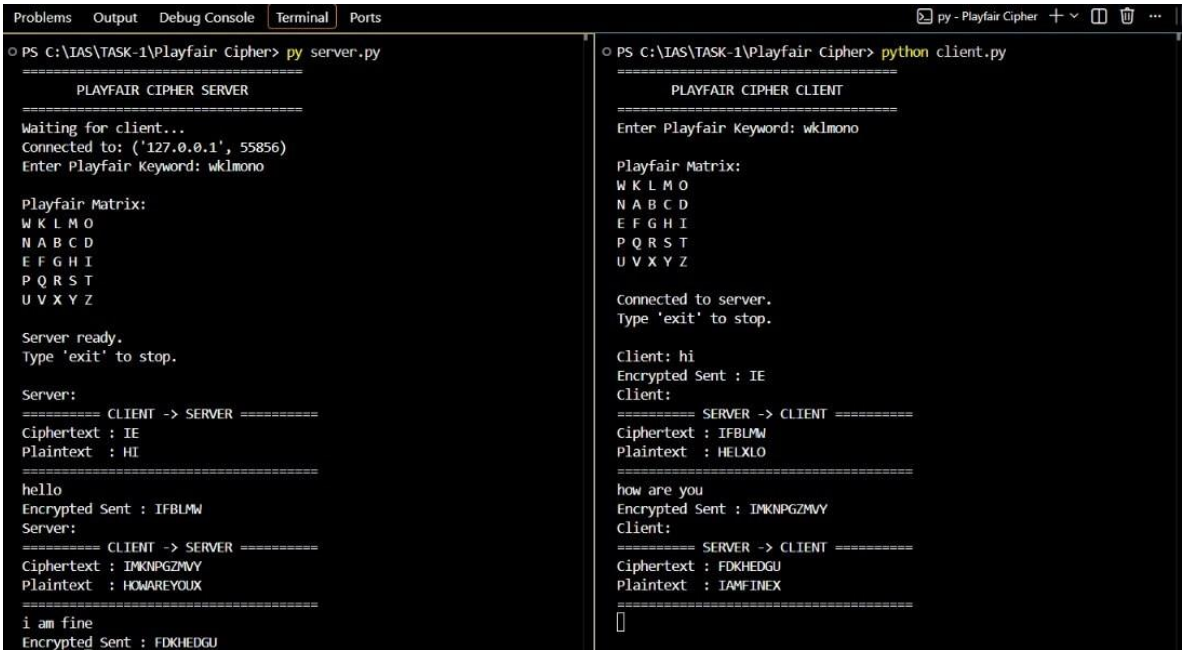

Figure 1.2: Playfair Cipher Client-Server Communication

1.3 Hill Cipher

1.3.1 Algorithm Definition

The Hill Cipher is a classical polygraphic substitution cipher based on matrix multiplication. A key matrix is used to transform blocks of plaintext characters into ciphertext.

1.3.2 Mathematical Notation

The encryption operation is:

$$C = KP \pmod{26} \tag{1.3}$$

The decryption operation is:

$$P = K^{-1}C \pmod{26} \tag{1.4}$$

where K is the key matrix, K^{-1} is the inverse key matrix, P is the plaintext vector, and C is the ciphertext vector.

For the inverse matrix to exist modulo 26:

$$\gcd(\det(K), 26) = 1 \tag{1.5}$$

1.3.3 Screenshot

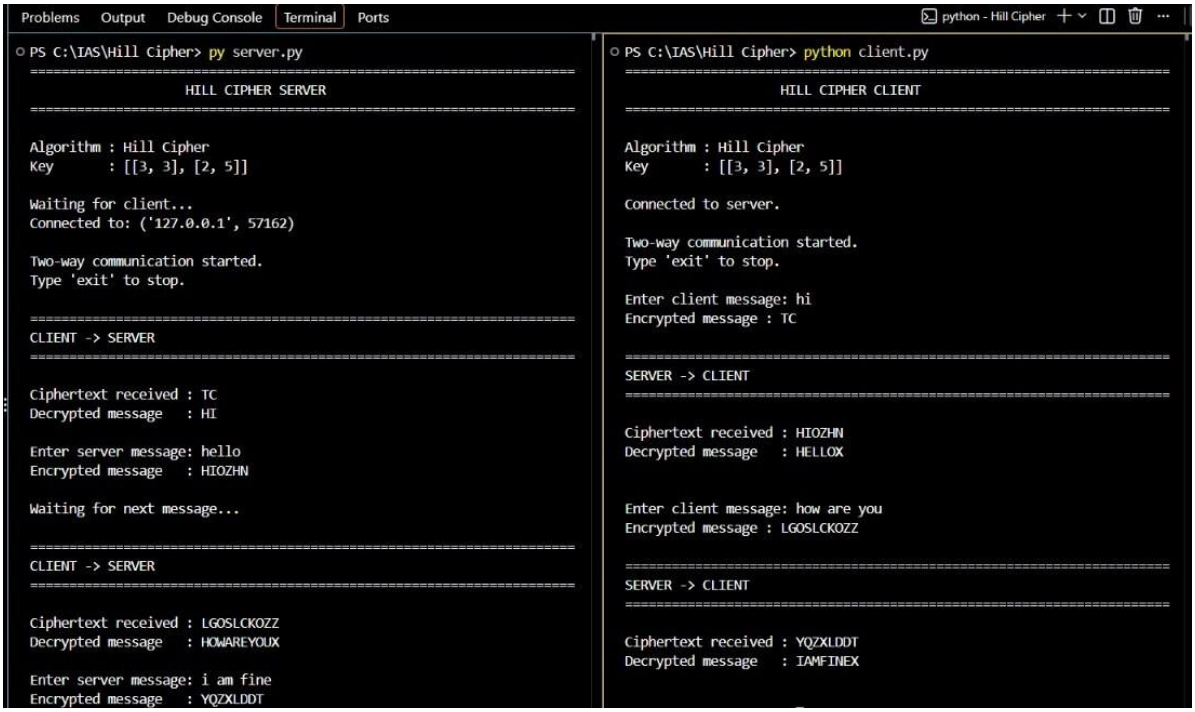

Figure 1.3: Hill Cipher Client-Server Communication

1.4 Vigenère Cipher

1.4.1 Algorithm Definition

The Vigenère Cipher is a polyalphabetic substitution cipher that uses a keyword to determine different shifts for successive plaintext characters.

1.4.2 Mathematical Notation

The encryption operation is:

$$C_i = (P_i + K_i) \bmod 26 \tag{1.6}$$

The decryption operation is:

$$P_i = (C_i - K_i) \bmod 26 \tag{1.7}$$

where K_i represents the corresponding key character.

1.4.3 Screenshot

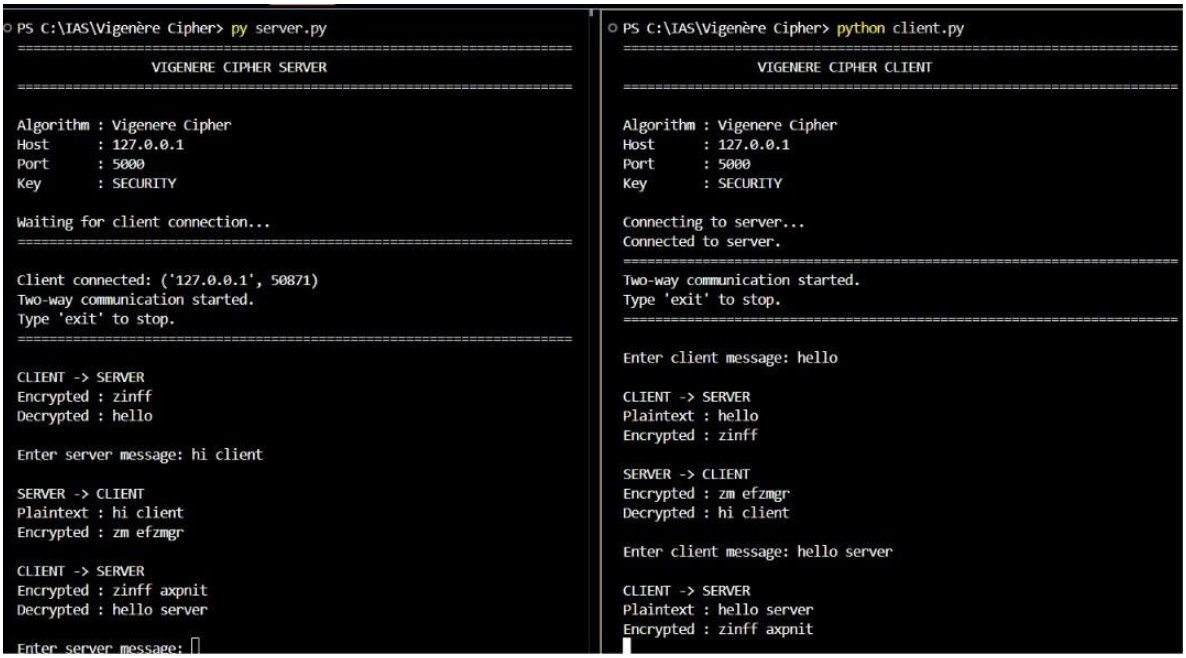

Figure 1.4: Vigenère Cipher Client-Server Communication

1.5 Simplified Data Encryption Standard (SDES)

1.5.1 Algorithm Definition

Simplified Data Encryption Standard (SDES) is an educational block cipher based on the concepts of permutations, substitutions, key generation, and Feistel-style processing.

SDES demonstrates the basic principles used in block cipher design, including key generation, permutation, substitution, and multiple rounds of encryption.

1.5.2 Mathematical Notation

The basic Feistel operation can be represented as:

$$L_{i+1} = R_i \tag{1.8}$$

$$R_{i+1} = L_i \oplus F(R_i, K_i) \tag{1.9}$$

where L_i and R_i represent the left and right portions of the data block, K_i represents the round key, and F represents the round function.

1.5.3 Screenshot

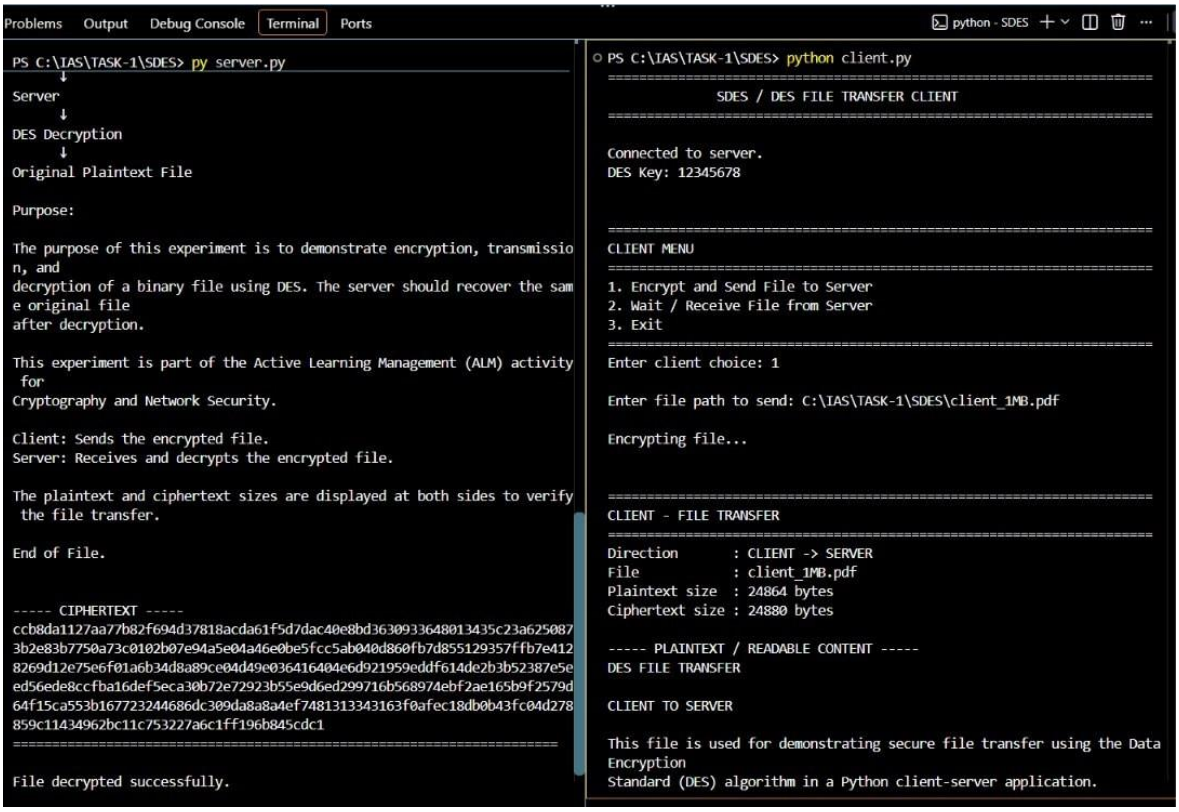

Figure 1.5: SDES Client-Server Communication

1.6 Advanced Encryption Standard (AES)

1.6.1 Algorithm Definition

Advanced Encryption Standard (AES) is a symmetric block cipher used for secure encryption of data. AES operates on 128-bit blocks and supports key sizes of 128, 192, and 256 bits.

The AES prototype uses AES for secure client-server file transfer. The implementation uses the PyCryptodome package.

```
1 from Crypto.Cipher import AES
```

Listing 1.1: PyCryptodome AES Import

1.6.2 Mathematical Notation

The main AES transformations are:

- SubBytes
- ShiftRows
- MixColumns
- AddRoundKey

A standard AES round can be represented as:

$$State' = AddRoundKey\ (MixColumns\ (ShiftRows\ (SubBytes(State)))) \quad (1.10)$$

The final AES round does not perform the MixColumns operation.

1.6.3 Screenshots

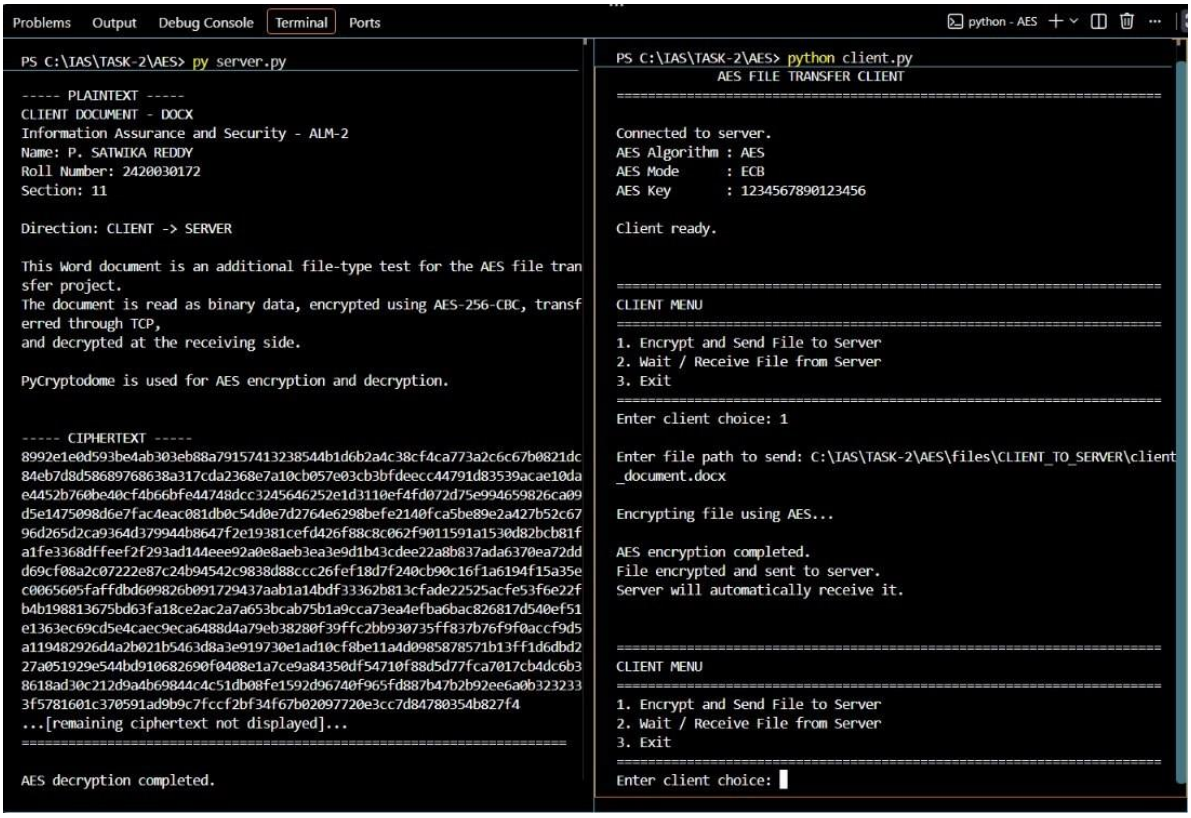

Figure 1.6: AES File Transfer from Client to Server

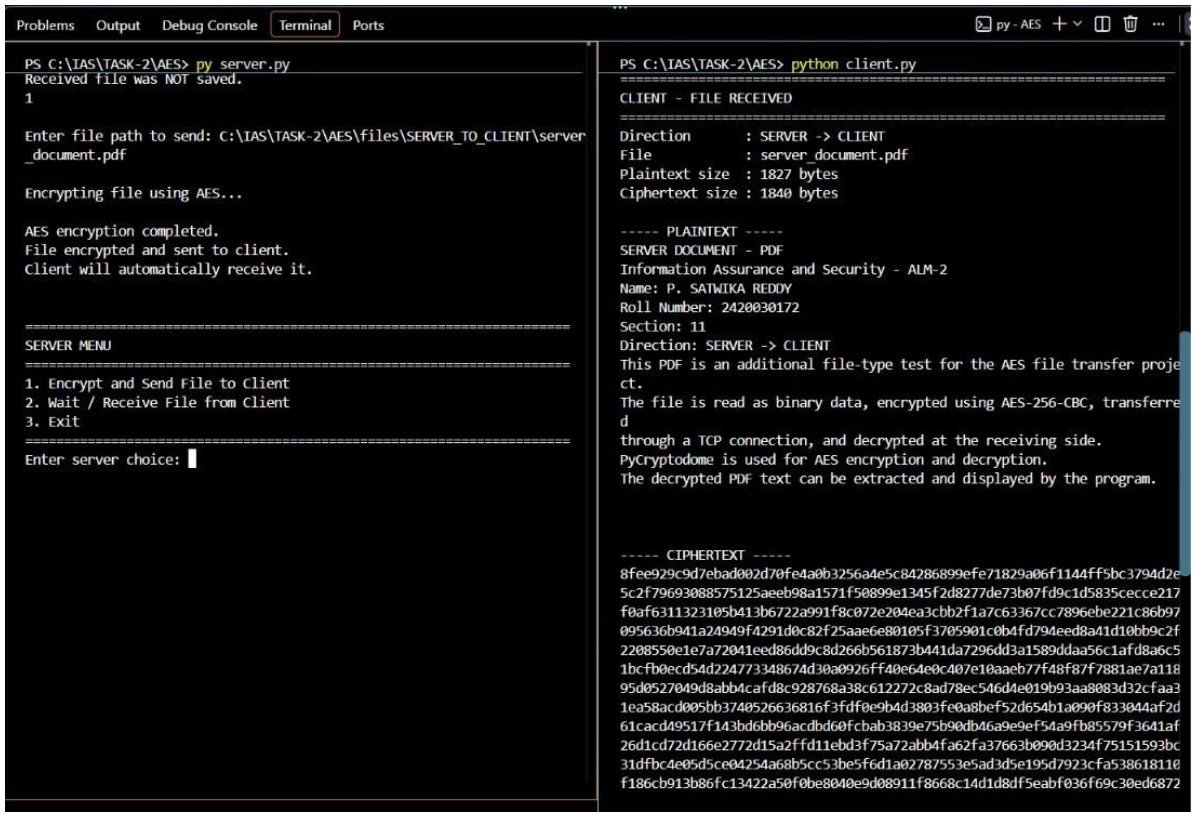

Figure 1.7: AES File Transfer from Server to Client

1.7 Stream Cipher (RC4)

1.7.1 Algorithm Definition

RC4 (Rivest Cipher 4) is a stream cipher that generates a pseudorandom keystream. The generated keystream is combined with the plaintext using the XOR operation.

The RC4 prototype uses the ARC4 implementation provided by PyCryptodome.

```
1 from Crypto.Cipher import ARC4
```

Listing 1.2: PyCryptodome RC4 Import

1.7.2 Mathematical Notation

The encryption operation is:

$$C_i = P_i \oplus K_i \tag{1.11}$$

The decryption operation is:

$$P_i = C_i \oplus K_i \tag{1.12}$$

where K_i represents the generated keystream.

The RC4 key scheduling algorithm generates the initial state, and the pseudo-random generation algorithm produces the keystream.

1.7.3 Screenshot

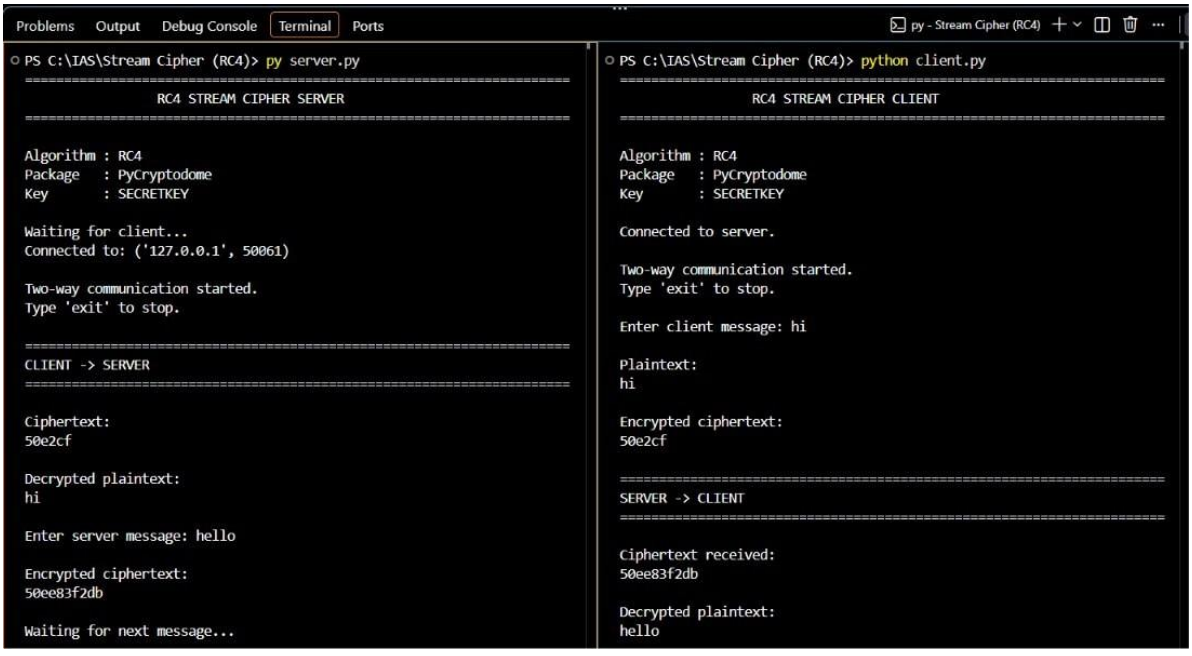

Figure 1.8: RC4 Client-Server Communication

1.8 Diffie–Hellman

1.8.1 Algorithm Definition

Diffie–Hellman is a key exchange algorithm that allows two parties to establish a shared secret over a communication channel without directly transmitting the secret key.

Both parties use common public parameters and their own private values to calculate the same shared secret.

1.8.2 Mathematical Notation

Let P be a public prime number and G be a public generator.

The client selects private value a and calculates:

$$A = G^a \bmod P \tag{1.13}$$

The server selects private value b and calculates:

$$B = G^b \bmod P \tag{1.14}$$

The client calculates the shared secret:

$$S = B^a \bmod P \tag{1.15}$$

The server calculates the shared secret:

$$S = A^b \bmod P \tag{1.16}$$

Therefore:

$$B^a \bmod P = A^b \bmod P \tag{1.17}$$

and both parties obtain the same shared secret.

1.8.3 Screenshot

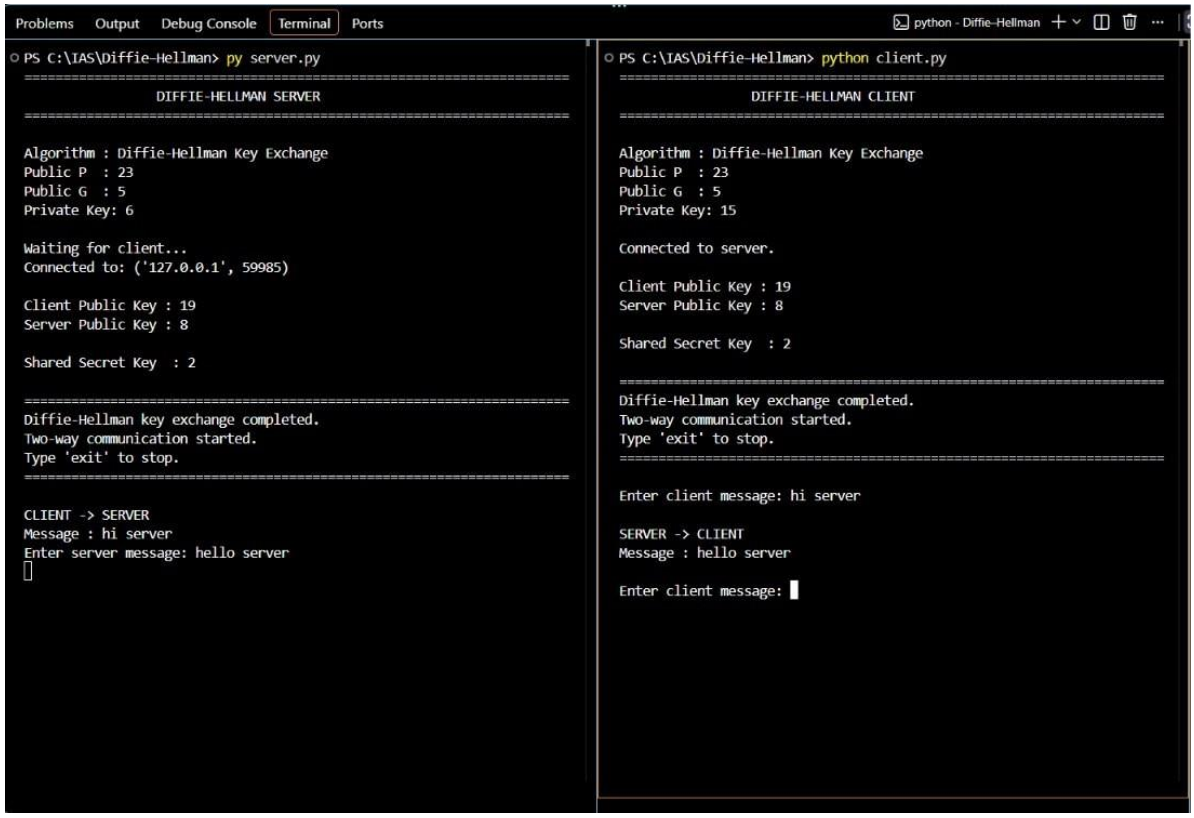

Figure 1.9: Diffie–Hellman Key Exchange and Communication

1.9 RSA

1.9.1 Algorithm Definition

RSA is an asymmetric cryptographic algorithm that uses a public key and a private key. The public key can be used for encryption, while the corresponding private key is used for decryption.

The RSA prototype uses the RSA functionality provided by PyCryptodome.

```
1 from Crypto.PublicKey import RSA
2 from Crypto.Cipher import PKCS1_OAEP
```

Listing 1.3: PyCryptodome RSA Imports

1.9.2 Mathematical Notation

Two prime numbers p and q are selected.

The modulus is calculated as:

$$n = pq \tag{1.18}$$

Euler's totient function is:

$$\phi(n) = (p - 1)(q - 1) \tag{1.19}$$

The public key is represented by (e, n) and the private key by (d, n).

The encryption operation is:

$$C = M^e \bmod n \tag{1.20}$$

The decryption operation is:

$$M = C^d \bmod n \tag{1.21}$$

where M is the plaintext message and C is the ciphertext.

1.9.3 Screenshot

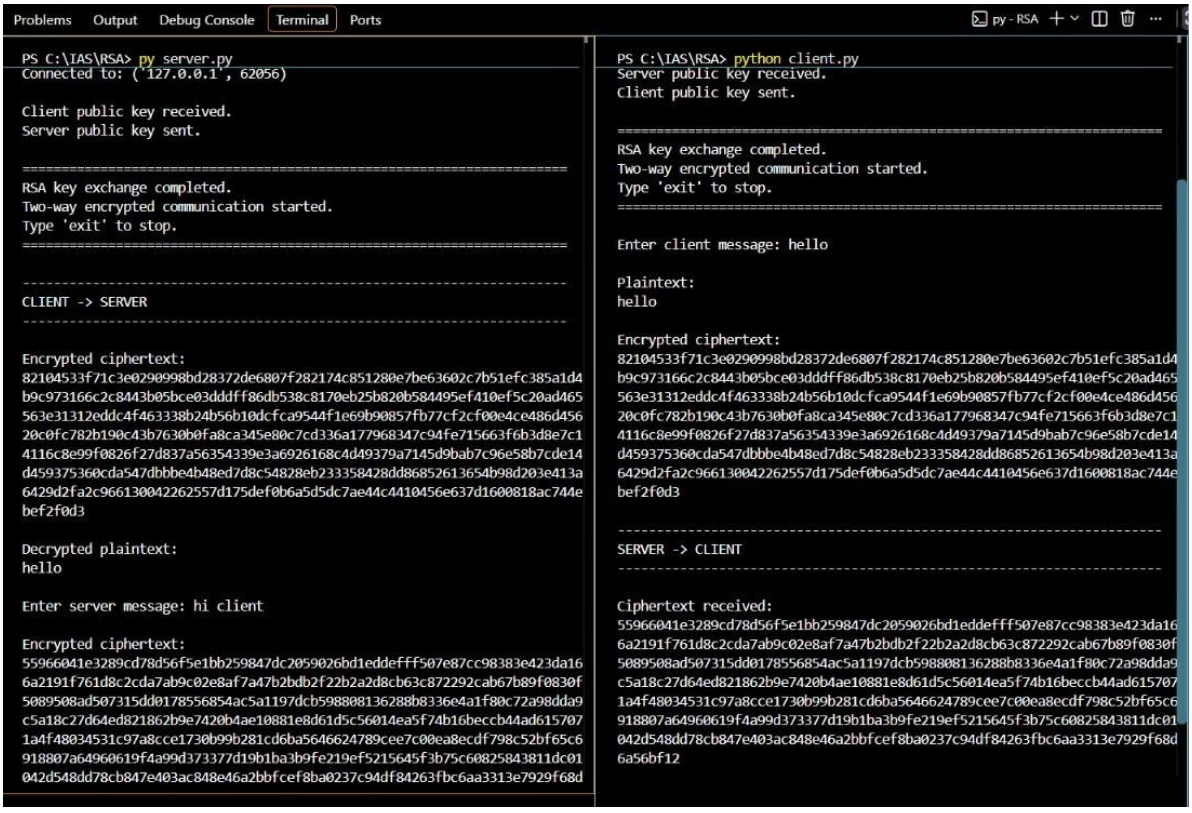

Figure 1.10: RSA Client-Server Communication

GitHub Repository

The complete implementation of the Information Assurance and Security prototype is available in the following GitHub repository:

https://github.com/satwika06bhaskar-design/IAS-2420030172.git